Uniqueness for spatially homogeneous rank-one-cone diffusions

Candidate full sufficient-condition theorem for arXiv:2605.00439

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Status. Candidate full theorem, likely valid, pending independent expert review. It gives weighted one-sided stability, comparison, and uniqueness for every bounded datum whenever the coefficient is a nonnegative combination of fixed rank-one directions with state-dependent weights.

1 Source question and result

Auscher and Bechtel consider bounded weak solutions on $\mathbb{R}^n$ of

$$\partial_t u - \operatorname{div}(a(t,x,u)\nabla u) = 0, \qquad u(0) = u_0. \tag{1}$$

Their Corollary 1.5 constructs a global bounded weak solution for every bounded datum, but their concluding uniqueness problem asks for suitable conditions ensuring uniqueness at this rough initial regularity [1].

Open problems. We conclude with some open problems:

- (Systems) Our Proposition 4.1 remains valid for systems, hence it would be interesting to look at quasilinear systems with our approach.
- (Uniqueness) For irregular initial data such as $u_0 \in \mathrm{L}^\infty(\mathbb{R}^n)$, it would be interesting to investigate suitable conditions ensuring uniqueness of bounded weak solutions. The issue is to have sufficient a priori regularity near the initial time.
- (Existence) It would be interesting to investigate whether the existence result in Corollary 1.5 can be extended to more irregular data such as $u_0 \in \mathrm{VMO}(\mathbb{R}^n)$.

Here is a broad structural answer. Fix nonzero vectors $e_1, \ldots, e_m \in \mathbb{R}^n$ spanning $\mathbb{R}^n$. Assume

$$a(t,x,s) = a(t,s) = \sum_{r=1}^{m} \alpha_r(t,s) e_r \otimes e_r, \qquad \alpha_r(t,s) > 0, \tag{2}$$

where each α_r is measurable in t, locally Lipschitz in s, and, on every bounded time interval and compact state interval, is bounded above and away from zero. Assume also that the resulting matrix a satisfies Assumption A of the source. Thus the weights may vary with time and the state while the elliptic directions remain fixed. Isotropic diffusion $a(t,s) = \alpha(t,s)\mathrm{I}$ and fixed-axis diagonal anisotropy are special cases.

For $p > n/2$, put

$$\omega_p(x) = (1 + |x|^2)^{-p}. \tag{3}$$

Theorem 1 (Weighted comparison and uniqueness)**.** *Assume* (2)*. Let u, v be bounded weak solutions of* (1)*, in the precise sense of [1, Definition 1.2], with bounded initial data u_0, v_0. For every $T < \infty$ there is $C_T < \infty$, depending only on p, the fixed directions, and the bounds of the α_r on the joint solution range up to time T, such that for almost every $t \in (0, T)$,*

$$\int_{\mathbb{R}^n} (u(t) - v(t))_+ \omega_p \leq e^{C_T t} \int_{\mathbb{R}^n} (u_0 - v_0)_+ \omega_p. \tag{4}$$

The analogous estimate holds for the negative part. Consequently:

1. *$u_0 \leq v_0$ almost everywhere implies $u \leq v$ almost everywhere;*
2. *two bounded weak solutions with the same bounded datum coincide;*
3. *under* (2)*, the solution supplied by source Corollary 1.5 is the unique global bounded weak solution.*

The argument has two ingredients. The fixed rank-one structure converts the difference equation into a sum of one-dimensional monotone diffusions. A directional Kato inequality then propagates a polynomially weighted L^1 distance. The source's own local L^2 trace theorem closes the estimate at $t = 0$, even though the solutions need not be integrable on $\mathbb{R}^n$.

2 Directional Kato lemma

Write $\mathrm{D}_e = e \cdot \nabla$. We use the standard parabolic Kato inequality in the following finite-direction form.

Lemma 2 (Finite-direction Kato inequality)**.** *Let $w, z_1, \ldots, z_m$ be locally integrable on $(0, T) \times \mathbb{R}^n$, with enough local energy regularity to justify the weak equation*

$$\partial_t w = \sum_{r=1}^{m} \mathrm{D}_{e_r}^2 z_r. \tag{5}$$

Suppose $z_r = q_r w$ with $q_r \in H^1_{\mathrm{loc}}$ locally bounded and strictly positive (locally bounded away from zero). Then, in distributions,

$$\partial_t (w)_+ \leq \sum_{r=1}^{m} \mathrm{D}_{e_r}^2 (z_r)_+. \tag{6}$$

The same assertion holds for negative parts and, after addition, for absolute values.

Proof. For smooth w, q_r, the claim is transparent. Since $q_r > 0$, $(z_r)_+ = q_r w_+$. Along a line parallel to e_r, the distributional chain rule gives

$$\mathrm{D}_{e_r}^2 (q_r w_+) - \mathbf{1}_{\{w>0\}} \mathrm{D}_{e_r}^2 (q_r w) = q_r\, \delta_{\{w=0\}} |\mathrm{D}_{e_r} w|^2 \geq 0;$$

all first and second derivatives of q_r cancel. Also $\partial_t w_+ = \mathbf{1}_{\{w>0\}} \partial_t w$. Multiplying (5) by this common indicator and summing the displayed inequalities proves (6).

For the stated weak regularity, take Steklov averages in time and smooth convex approximations j_ε of $s \mapsto s_+$, perform the same calculation after spatial mollification, and pass first in the mollifier, then the Steklov scale, and finally ε. The H^1_{loc} regularity of q_r, its local bounds, and the local energy bounds control the commutators; the only surviving convexity term is the nonnegative measure displayed above. This is the standard parabolic Kato approximation, applied separately in each fixed direction and then summed. □

For the solutions considered below, u, v and their spatial gradients are locally square integrable, and every nonlinear primitive is locally H^1 by the source's local Lipschitz assumption. Thus Lemma 2 applies by the displayed approximation argument. This is also the familiar L^1-accretivity mechanism for scalar nonlinear diffusion; compare the bounded, non-integrable entropy theory in [2].

3 Difference equation and weighted estimate

Fix $s_* \in O$. For each direction define the increasing primitive

$$\beta_r(t,s) = \int_{s_*}^{s} \alpha_r(t,q)\,dq. \tag{7}$$

Since α_r is independent of x, the Sobolev chain rule gives

$$\operatorname{div}(a(t,u)\nabla u) = \sum_{r=1}^{m} \mathrm{D}_{e_r}^2 \beta_r(t,u)$$

in distributions. Set

$$w = u - v, \qquad z_r = \beta_r(t,u) - \beta_r(t,v).$$

More explicitly,

$$z_r = q_r w, \qquad q_r(t,x) = \int_0^1 \alpha_r(t, v(t,x) + \theta w(t,x))\,d\theta.$$

The local Lipschitz assumption and $u, v \in H^1_{\mathrm{loc}}$ give $q_r \in H^1_{\mathrm{loc}}$ in space. Subtracting the weak equations gives (5). Because every α_r is positive,

$$\operatorname{sgn} z_r = \operatorname{sgn} w \quad (z_r \neq 0), \qquad |z_r| \leq L_{r,T}|w|, \tag{8}$$

where $L_{r,T}$ is the supremum of α_r on the compact joint range of u, v and on $[0,T]$. Lemma 2 therefore yields

$$\partial_t w_+ \leq \sum_r \mathrm{D}_{e_r}^2 (z_r)_+. \tag{9}$$

The weight (3) is integrable because $p > n/2$, and direct differentiation gives

$$|\mathrm{D}_e^2 \omega_p(x)| \leq C_p |e|^2 \omega_p(x), \qquad C_p = 2p + 4p(p+1). \tag{10}$$

Test (9) against increasingly large compactly supported approximations to ω_p. The cutoff errors vanish by boundedness of z_r and integrability of ω_p and its first two derivatives. Using (8)–(10), in the sense of distributions in time,

$$\frac{d}{dt}\int w_+\omega_p \leq \sum_r \int (z_r)_+ \mathrm{D}_{e_r}^2 \omega_p \tag{11}$$

$$\leq C_p \sum_r L_{r,T}|e_r|^2 \int w_+\omega_p = C_T \int w_+\omega_p. \tag{12}$$

It remains only to identify the value at time zero. A bounded quasilinear solution is also a bounded weak solution of the linear equation with coefficient $A(t,x) = a(t, u(t,x))$. On the compact solution range, A is bounded and uniformly elliptic. Hence [1, Proposition 3.4(b)] gives

$$u(t) \to u_0, \qquad v(t) \to v_0 \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}^n) \quad (t \downarrow 0). \tag{13}$$

The weighted L^1 convergence follows: on a fixed ball use (13) and Cauchy–Schwarz, while outside the ball use the uniform bounds on u, v, u_0, v_0 and then send the radius to infinity because $\omega_p \in L^1$. Therefore

$$\int (u(t) - v(t))_+ \omega_p \longrightarrow \int (u_0 - v_0)_+ \omega_p.$$

Gronwall's lemma applied to (11) proves (4). Reversing u, v proves the negative-part estimate, and all conclusions of Theorem 1 follow.

4 Examples, boundary of the method, and novelty audit

The theorem includes

$$a(t,s) = \alpha(t,s)\mathrm{I}, \qquad a(t,s) = \operatorname{diag}(\alpha_1(t,s), \ldots, \alpha_n(t,s)),$$

and overcomplete fixed frames $a(t,s) = \sum_r \alpha_r(t,s) e_r \otimes e_r$. The weights need not coincide, and may be time dependent. The source permits local degeneracy as $|s| \to \infty$; only uniform ellipticity on the bounded joint solution range is needed.

The proof does not cover rotating eigendirections depending on s, nor x-dependent coefficients: differentiating a state primitive then creates spatial drift terms, and sign-separated directional Kato no longer applies directly. Those are genuine remaining parts of the source's broad uniqueness program.

Exact-title, exact-equation, Kato, nonlinear-diffusion, and rank-one-cone searches through 11 August 2026 found classical isotropic entropy and L^1-contraction results, notably [2], but no statement matching the fixed-rank-one cone for the source's bounded weak-solution class and rough distributional trace. The present argument is self-contained modulo the standard Kato approximation detailed in Lemma 2 and the source's own local L^2 trace theorem.

One final triage point is worth recording. The source's VMO existence bullet cannot literally enlarge its *bounded* weak-solution class to unbounded VMO traces: Proposition 3.4(b) already forces every bounded solution trace to lie in L^∞. A separate supporting note in this packet's companion attempt gives an explicit locally bounded unbounded CMO datum. The rank-one-cone uniqueness theorem above is the genuinely live result.

References

[1] P. Auscher and S. Bechtel, *Existence and uniqueness of weak solutions to quasilinear PDEs with critical data*, arXiv:2605.00439v3, 2026.

[2] J. Endal and E. R. Jakobsen, *L^1 contraction for bounded (non-integrable) solutions of degenerate parabolic equations*, SIAM J. Math. Anal. **46** (2014), 3957–3982; arXiv:1404.6418v2.